

Atharva Hatolkar

Data Scientist · Machine Learning Engineer

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PROFESSIONAL SUMMARY

Data scientist and machine learning engineer completing an MSc in Data Analytics at the University of Galway, with 2.5 years of professional experience in data engineering, SQL analytics, and business intelligence. Specialises in time-series forecasting, deep learning, and statistical model evaluation, with hands-on experience building and benchmarking transformer and foundation models on large-scale, real-world clinical datasets. Comfortable across the full data workflow — data extraction and cleaning, feature engineering, predictive modelling, evaluation, and communicating results to technical and non-technical stakeholders. Available from August 2026; eligible to work in Ireland with no employer sponsorship required.

CORE COMPETENCIES

Machine Learning · Deep Learning · Time-Series Forecasting · Predictive Modelling · Statistical Analysis · Feature Engineering · Exploratory Data Analysis · Data Wrangling · Model Evaluation · ETL & Data Pipelines · Business Intelligence · Data Visualization · Stakeholder Communication

TECHNICAL SKILLS

Programming: Python, SQL, PL/SQL, R, Bash

Machine Learning & AI: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, deep learning, time-series forecasting, foundation models, feature engineering, model evaluation & validation

Data & Cloud: Google BigQuery, OMOP Common Data Model, pandas, NumPy, Jupyter, Git / GitHub, Linux

Analytics & Visualization: Power BI, Tableau, Excel, Matplotlib, Seaborn, Plotly

Statistics & Methods: regression, hypothesis testing, probability, cross-validation, experimental evaluation

PROFESSIONAL EXPERIENCE

Senior Systems Engineer — Data & Integration

April 2023 – August 2025

Trinamix Systems — Pune, India

- Built and optimised SQL and PL/SQL queries and data pipelines to extract, transform, and prepare large datasets from enterprise systems for analysis and reporting.
- Designed automated data-quality, validation, and reconciliation routines that improved the accuracy and reliability of downstream analytics and reporting datasets.
- Developed interactive business-intelligence dashboards and reports in Power BI and Tableau, turning complex operational data into clear, decision-ready insight for cross-functional stakeholders.
- Performed exploratory data analysis on large relational datasets to identify trends, anomalies, and data-integrity issues, surfacing findings that informed business decisions.
- Collaborated with global teams across the US, EMEA, and APAC to translate ambiguous business questions into data specifications and deliver analytical solutions on schedule.
- Owned end-to-end delivery of recurring reporting and data-integration workflows, ensuring consistency and reliability across multiple enterprise clients.

- Promoted from Systems Engineer to Senior Systems Engineer within 18 months in recognition of consistent delivery, analytical rigour, and technical ownership.

PROJECTS

MSc Dissertation — Forecasting ICU Vital Signs with Time-Series Foundation Models

2026

University of Galway · Supervised by Prof. Michael Madden

- Benchmarked transformer models (iTransformer, GraFITi) and pretrained time-series foundation models (Chronos-2, TimesFM, Moirai-2) to forecast 13 physiological signals during ventilator weaning for over 1,600 intensive-care patients.
- Built the patient cohort from the AmsterdamUMCdb critical-care database, standardised it to the OMOP Common Data Model, and queried it at scale using Google BigQuery.
- Engineered a leakage-safe preprocessing pipeline handling irregular sampling, per-patient train / validation / test splits, imputation with observation masking, and windowing at +1h, +4h, and +8h forecast horizons.
- Structured the benchmark in tiers — transformers trained from scratch, zero-shot foundation models, and irregularity-native graph models — to isolate the effect of pretraining and architecture on forecast accuracy.
- Analysed cross-variable attention in the transformer models to interpret which physiological signals drove each forecast, separating genuine signal from imputation artefacts.
- Evaluated models with MAE, RMSE, and MASE against a persistence baseline, stratified by forecast horizon and physiological variable for fair cross-model comparison.

Result: GraFITi achieved the strongest accuracy, leading on 11 of 13 signals with a mean MASE of 0.79 — outperforming iTransformer (0.82) and Chronos-2 (0.84).

Tools: Python, PyTorch, Hugging Face, Google BigQuery, pandas, NumPy

Code: github.com/HatolkarAV/ICU-Weaning-Prediction

EDUCATION

MSc, Computer Science (Data Analytics)

Sept 2025 – Aug 2026 (expected)

University of Galway, Ireland

Relevant coursework: machine learning, deep learning, natural language processing, statistics, optimisation, and large-scale data analytics.

PG Diploma, Big Data Analytics

2022 – 2023

Centre for Development of Advanced Computing (CDAC), Pune, India

Applied big-data processing, data engineering, machine learning, and analytics.

B.Tech, Electronics & Telecommunications Engineering

2018 – 2022

Government College of Engineering, Jalgaon, India

Foundation in signal processing, networks, probability, and programming.

LANGUAGES

English (professional working proficiency) · Hindi (native) · Marathi (native)

CERTIFICATIONS & ADDITIONAL

- Google Cloud and Oracle certifications.
- Competitive achievements: district-level chess and state-level kabaddi.
- Work eligibility: Ireland (Stamp 1G graduate permission on completion; no employer sponsorship required).